Copper(I) Iodide Mediated Synthesis of Carbohydrate-Based Terminal Allenes by ATA Reaction

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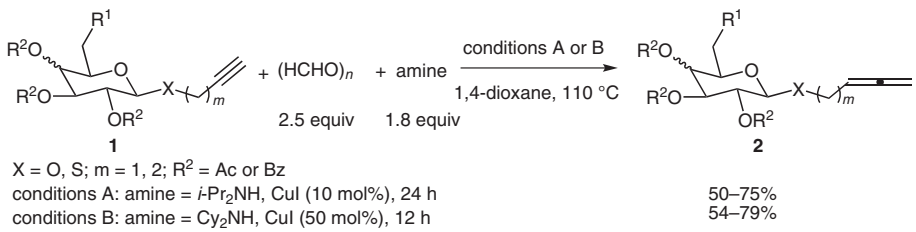
Abstract: We have demonstrated two improved methods for the synthesis of carbohydrate-based terminal allenes from propargyl or homopropargyl glycosides. The reaction of propargyl or homopropargyl glycosides with paraformaldehyde (2.5 equiv) in the presence of copper(I) iodide (10 mol%) and diisopropylamine (1.8 equiv) or copper(I) iodide (50 mol%) and dicyclohexylamine (1.8 equiv) produces sugar-linked terminal allenes in 50–79% yield.

Key words: copper(I) iodide, diisopropylamine, dicyclohexylamine, carbohydrates, allenes

Carbohydrate chemistry has become more and more important, since carbohydrates exist widely in natural products with remarkable biological activities.[1] However, there are limited reports[2] on allenes with carbohydrate moieties. Sugar-linked 1,2-allenyl ethers are usually prepared by the isomerization of propargyl glycosides in the presence of a strong base.[2a–c] During the past few decades, allene chemistry has experienced great progress, since allenes are very useful intermediates in organic synthesis.[3] About 150 natural products containing an allenic or cumulenic structure have been found, and this turned out to be a very attractive and prolific area of interest.[4] So far, different kinds of synthetic methods to produce functional allenes[5] have been developed, among which the allenylation of terminal alkynes (ATA reaction) with paraformaldehyde [$(HCHO)_n$] in the presence of amines and copper(I) salts may be the most straightforward method to obtain terminal allenes.[6] Although many functional groups such as mesylate, ether, hydroxy, amide, and so forth are tolerated under the current conditions, it is still interesting to introduce this method for the synthesis of more complicated carbohydrate-based allenes, since some functionalities are not well tolerated under the original protocol.[7] In this paper, we wish to report an easy application of copper(I) iodide mediated synthesis of carbohydrate-based terminal allenes from propargyl or homopropargyl glycosides (Scheme 1).

Initially, we chose prop-2-ynyl-substituted 2,3,4,6-tetra-*O*-acetyl-β-D-glucopyranoside **1a**[8] as the model substrate in search for the best reaction conditions. Under the conditions of our previous report,[6f] the reaction of **1a** (2.0 mmol) with paraformaldehyde (1.6 equiv), diisopropylamine (1.4 equiv), and copper(I) iodide (10 mol%) in 1,4-dioxane at 110 °C afforded allenyl-substituted glucopyranoside **2a** in 55% yield (Table 1, entry 1). Fortunately, increasing the amount of aldehyde and amine to 2.5 equivalents and 1.8 equivalents, respectively, gave a higher yield of 63% (entry 2). However, replacing copper(I) iodide with copper(I) bromide[6a] or diisopropylamine with dicyclohexylamine led to lower yields (entries 3 and 4). To our delight, when 50 mol% copper(I) iodide was used, the yield was improved to 70% (entry 5). Furthermore, dicyclohexylamine[6d] seems to be slightly better than diisopropylamine with copper(I) iodide (50 mol%), affording **2a** in 73% yield (entry 6). Thus, we defined the combination of copper(I) iodide (10 mol%) and diisopropylamine (1.8 equiv) as standard conditions A and the combination of copper(I) iodide (50 mol%) and dicyclohexylamine (1.8 equiv) as standard conditions B.

With the standard conditions A and B in hand, the substrate scope and generality of the reaction were investigated and the results are summarized in Table 2. For propargyl-substituted β-D-glucopyranosides[8] **1a–d** bearing acetoxy, tosyloxy, octanoylamino, and bis(methoxy-

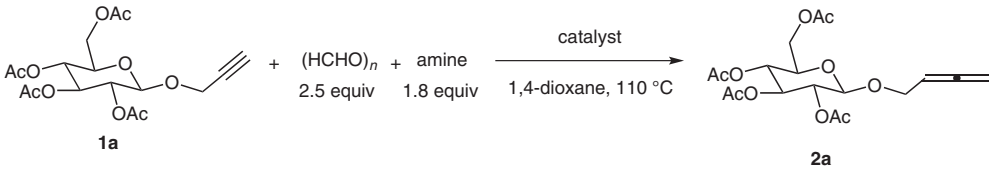


Scheme 1 Copper(I) iodide mediated synthesis of carbohydrate-based terminal allenes

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Table 1 Optimization of Conditions for Reaction of **1a** with Paraformaldehyde, Amine, and Copper(I) Catalyst[a]

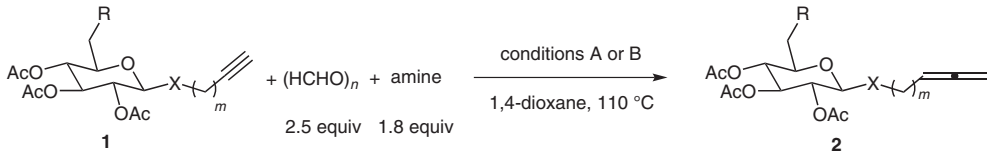


Entry	Catalyst (mol%)	Amine	Time (h)	Isolated yield (%)
1[b]	CuI (10)	*i*-Pr_2NH	24	55
2	CuI (10)	*i*-Pr_2NH	24	63
3	CuBr (10)	*i*-Pr_2NH	24	48
4	CuI (10)	Cy_2NH	24	54
5	CuI (50)	*i*-Pr_2NH	12	70
6	CuI (50)	Cy_2NH	12	73

[a] Reaction conditions: **1a** (2.0 mmol), $(HCHO)_n$ (2.5 equiv), amine (1.8 equiv), 1,4-dioxane (3.0 mL), 110 °C.
[b] $(HCHO)_n$ (1.6 equiv) and *i*-Pr_2NH (1.4 equiv) were used.

carbonyl)methyl at the C-6 position, respectively, the reaction afforded the corresponding allenyl β-D-glucopyranosides **2a**–**d** in 50–72% isolated yields under both standard conditions. Except for **2c**, the yields of products **2** are higher under conditions B than under conditions A (Table 2, entries 1–4). Similarly, for homopropargyl-substituted β-D-glucopyranosides[9] **1e**–**g**, the expected products **2e**–**g** were obtained in 51–67% yields and the yields under conditions B are slightly higher than those under conditions A (entries 4–7). Propargyl-substituted β-D-thioglucoside[10] **1h** also provided the corresponding product **2h** in 71% yield under both conditions (entry 8). The structure of **2e** was further established by an X-ray diffraction study (Figure 1).[11]

Table 2 The Copper(I) Iodide Mediated Reaction of Substrates **1**, Paraformaldehyde, and Diisopropylamine or Dicyclohexylamine in 1,4-Dioxane[a]



X = O, S; m = 1, 2
conditions A: amine = *i*-Pr_2NH, CuI (10 mol%), 24 h
conditions B: amine = Cy_2NH, CuI (50 mol%), 12 h

Entry	Substrates **1**			Conditions A		Conditions B	
	R	X	*m*	Scale (mmol)	Yield of **2** (%)[b]	Scale (mmol)	Yield of **2** (%)[b]
1	OAc	O	1 (**1a**)	10	64	10	72 (**2a**)
2	OTs	O	1 (**1b**)	10	52	10	54 (**2b**)
3	$NHCOC_7H_{15}$	O	1 (**1c**)	5	64	5	55 (**2c**)
4	$CH(CO_2Me)_2$	O	1 (**1d**)	3	50	3	59 (**2d**)
5	OAc	O	2 (**1e**)	10	51	10	66 (**2e**)
6[c]	OTs	O	2 (**1f**)	5	52	5	54 (**2f**)
7	$CH(CO_2Me)_2$	O	2 (**1g**)	3	59	3	67 (**2g**)
8	OAc	S	1 (**1h**)	10	71	12	71 (**2h**)

[a] Reaction conditions A: **1** (0.67 M in 1,4-dioxane), CuI (10 mol%), $(HCHO)_n$ (2.5 equiv), *i*-Pr_2NH (1.8 equiv), 110 °C, 24 h; Reaction conditions B: **1** (0.67 M in 1,4-dioxane), CuI (50 mol%), $(HCHO)_n$ (2.5 equiv), Cy_2NH (1.8 equiv), 110 °C, 12 h.
[b] Isolated yield.
[c] The reaction time for conditions B was 13 h.

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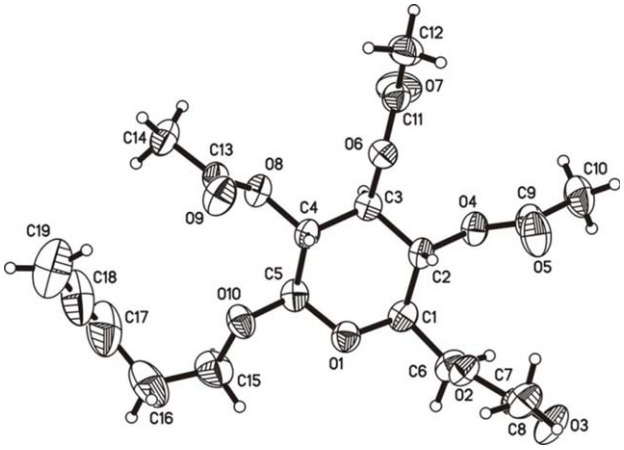
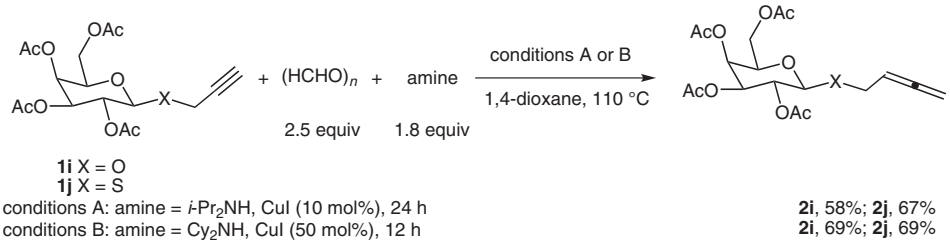


Figure 1 ORTEP representation of **2e**

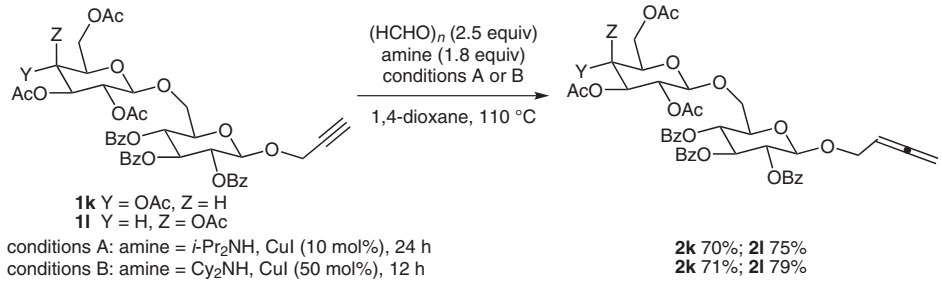
In addition, when propargyl-substituted galactopyranoside[8] and propargyl-substituted thiogalactopyranoside[10] **1i** and **1j** were used, the corresponding products **2i** and **2j** were also afforded in moderate yields of 58–69% (Scheme 2). To our delight, these two protocols were also suitable for the synthesis of allenyl disaccharides[12] **2k** and **2l** in good yields of 70–79% (Scheme 3).

Interestingly, when propargyl 2,3,4,6-tetra-*O*-propargyl-β-D-glucopyranoside[13] **1m** was used, all five terminal alkyne units could be transformed into terminal allene units under both conditions A and B, providing **2m** in an isolated yield of 58% and 61%, respectively (Scheme 4).

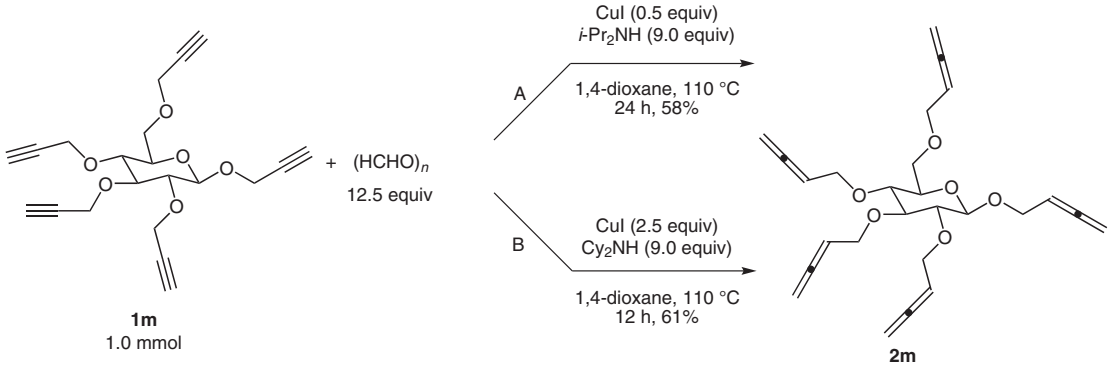
In conclusion, we have successfully demonstrated two improved methods of allenylation of terminal alkynes developed in this group to synthesize terminal 2,3-allenyl glycosides in moderate to good yields, which could tolerate functional groups such as ether, ester, amide, and malonate units. Allenyl β-D-glucopyranosides, β-D-galactopyranoside, thioglycosides, and disaccharides, and even pentakis-allenyl β-D-glucopyranoside could all be synthesized with these two methods. Further applications



Scheme 2 Synthesis of allenyl (thio)galactopyranosides **2i** and **2j**



Scheme 3 Synthesis of allenyl disaccharides **2k** and **2l**



Scheme 4 Synthesis of pentakis-allenyl β-D-glucopyranoside **2m**

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of these carbohydrate-based allenes are being pursued in our laboratory.

^{1}H and ^{13}C NMR spectra were recorded on an instrument operated at 300 MHz for ^{1}H NMR and 75 MHz for ^{13}C NMR. $CDCl_3$ was used as solvent in all NMR experiments. IR spectra of liquids were recorded of thin films of pure samples on NaCl plates, and for solid samples, KBr discs were used. Mass and HRMS spectra were carried out in ESI mode. TLC was performed on pre-coated glass-backed plates and UV light at 254 nm was used for visualization. Flash column chromatography was performed on silica gel. 1,4-Dioxane and paraformaldehyde [$(HCHO)_n$] were bought from Sinopharm Chemical Reagent Co., Ltd. CuI, *i*-Pr_2NH, and Cy_2NH were bought from Aladdin Industrial Corporation.

β-D-Glucopyranoside 1c (Scheme 5)[8c]

A solution of **3**[8b] (12.9168 g, 35.0 mmol) in CH_2Cl_2 (150 mL) and octanoyl chloride (8.4 mL, *d* = 0.953 g/mL, 8.0052 g, 49.0 mmol) were added to a flame-dried three-neck flask equipped with a dropping funnel. Then a solution of Ph_3P (12.8394 g, 49.0 mmol) in CH_2Cl_2 (30 mL) was added dropwise for 15 min, and the resulting solution was stirred for 48 h at r.t. under a N_2 atmosphere. After the reaction was complete as monitored by TLC (PE–EtOAc, 2:1), the resulting mixture was quenched with brine (50 mL) and extracted with CH_2Cl_2 (2 × 50 mL). The combined organic layer was washed with H_2O (100 mL) and brine (100 mL), and dried (Na_2SO_4). Filtration, evaporation of the solvent, and chromatography (silica gel, PE–EtOAc, 2:1) afforded **1c** as a syrup.

Yield: 8.4737 g (52%); $[\alpha]_D^{20}$ –27.6 (*c* = 0.897, $CHCl_3$).

IR (neat): 3399, 3284, 2959, 2929, 2857, 2117, 1756, 1652, 1538, 1431, 1368, 1246, 1220, 1164, 1061 cm^{-1}.

^{1}H NMR (300 MHz, $CDCl_3$): δ = 5.91 (t, *J* = 5.4 Hz, 1 H, NH), 5.23 (t, *J* = 9.5 Hz, 1 H), 5.04–4.86 (m, 2 H), 4.76 (d, *J* = 8.1 Hz, 1 H), 4.46–4.28 (m, 2 H), 3.70–3.52 (m, 2 H), 3.50–3.35 (m, 1 H), 2.55–2.48 (m, 1 H), 2.24–2.14 (m, 2 H, CH_2), 2.06 (s, 6 H, Me × 2), 2.00 (s, 3 H, Me), 1.70–1.55 (m, 2 H, CH_2), 1.40–1.18 (m, 8 H, CH_2 × 4), 0.88 (t, *J* = 6.2 Hz, 3 H, Me).

^{13}C NMR (75 Hz, $CDCl_3$): δ = 173.1, 169.9, 169.4, 169.2, 98.2, 78.0, 75.4, 72.3, 70.7, 68.8, 55.9, 38.7, 36.2, 31.4, 28.9, 28.7, 25.3, 22.3, 20.34, 20.27, 13.8.

ESI-MS: *m*/*z* = 492 [M^+ + Na], 470 [M^+ + H].

HRMS: *m*/*z* calcd for $C_{23}H_{36}N_1O_9$ [M^+ + H]: 470.2385; found: 470.2389.

β-D-Glucopyranoside 1d (Scheme 6)

To a three-necked flask were added sequentially **1b** (14.9512 g, 30.0 mmol), K_2CO_3 (16.5673 g, 120.0 mmol), DMSO (150 mL), and dimethyl malonate (17.2 mL, *d* = 1.15 g/mL, 19.8 g, 150.0 mmol). The reaction was complete after the mixture had stirred at 40 °C for 4 d as monitored by TLC (PE–EtOAc, 2:1). The resulting mixture was then slowly poured into a 1000 mL flask, and neutralized with 6 M HCl. The mixture was further extracted with CH_2Cl_2 (3 × 150 mL), washed with H_2O (250 mL) and brine (250 mL), and dried (Na_2SO_4). Filtration, evaporation of the solvent, and chromatography (silica gel, PE–EtOAc, 2.5:1 to 2:1) afforded crude **1d**, which was recrystallized from anhydrous EtOH; this gave pure **1d**.

Yield: 5.5364 g (40%); solid; mp 78–80 °C (anhydrous EtOH); $[\alpha]_D^{20}$ –20.3 (*c* = 0.857, $CHCl_3$).

IR (KBr): 3283, 2956, 2887, 2117, 1752, 1438, 1377, 1341, 1248, 1224, 1160, 1051 cm^{-1}.

^{1}H NMR (300 MHz, $CDCl_3$): δ = 5.20 (t, *J* = 9.5 Hz, 1 H), 4.97 (dd, *J* = 9.8, 8.0 Hz, 1 H), 4.90 (t, *J* = 9.6 Hz, 1 H), 4.67 (d, *J* = 8.1 Hz, 1 H), 4.34 (dd, *J* = 2.4, 0.3 Hz, 2 H), 3.78 (s, 3 H, Me), 3.74 (s, 3 H, Me), 3.69 (dd, *J* = 9.6, 5.3 Hz, 1 H), 3.57 (td, *J* = 9.9, 3.0 Hz, 1 H), 2.49 (t, *J* = 2.4 Hz, 1 H), 2.32–2.21 (m, 1 H), 2.13–2.03 (m, 1 H), 2.06 (s, 3 H, Me), 2.05 (s, 3 H, Me), 2.01 (s, 3 H, Me).

^{13}C NMR (75 Hz, $CDCl_3$): δ = 170.1, 169.6, 169.4, 169.2, 169.1, 98.2, 78.1, 75.4, 72.6, 71.8, 71.1, 70.9, 55.9, 52.68, 52.66, 47.5, 30.3, 20.6, 20.5.

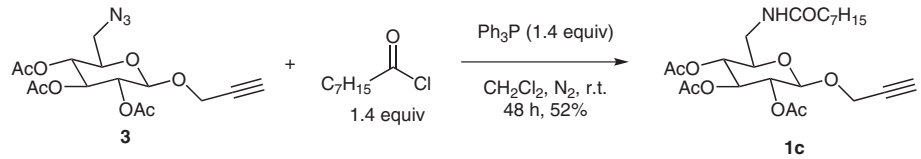
ESI-MS: *m*/*z* = 481 [M^+ + Na], 476 [M^+ + NH_4].

Anal. Calcd for $C_{20}H_{26}O_{12}$: C, 52.40; H, 5.72. Found: C, 52.49; H, 5.70.

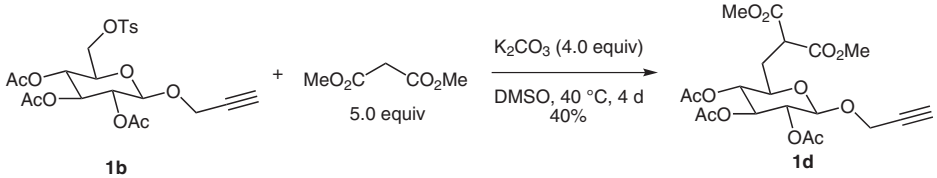
β-D-Glucopyranoside 1g (Scheme 7)

To a three-necked flask equipped with a reflux condenser were added sequentially **1f** (23.0427 g, 45.0 mmol), K_2CO_3 (24.8432 g, 180.0 mmol), DMSO (180 mL), and dimethyl malonate (25.8 mL, *d* = 1.15 g/mL, 29.7 g, 225.0 mmol). The reaction was complete after the mixture had stirred at 50 °C for 3 d as monitored by TLC (PE–EtOAc, 2:1). The resulting mixture was then slowly poured into a 1000 mL flask and neutralized with 6 M HCl. The mixture was further extracted with CH_2Cl_2 (3 × 200 mL), washed with H_2O (300 mL) and brine (300 mL), and dried (Na_2SO_4). Filtration, evaporation of the solvent, and chromatography (silica gel, PE–EtOAc, 2:1 to 1:1) afforded crude **1g**, which was recrystallized (EtOAc–hexane); this gave pure **1g**.

Yield: 8.4323 g (40%); solid; mp 107–108 °C (EtOAc–hexane); $[\alpha]_D^{20}$ +3.0 (*c* = 1.083, $CHCl_3$).



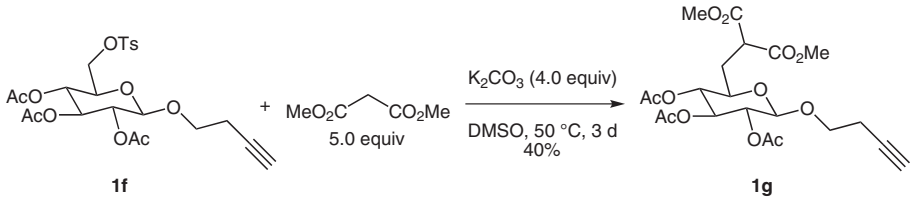
Scheme 5



Scheme 6

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Scheme 7

IR (KBr): 3291, 2965, 2920, 2866, 2117, 1757, 1439, 1376, 1338, 1242, 1214, 1171, 1078, 1060, 1032, 1010 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.17 (t, *J* = 9.6 Hz, 1 H), 4.95 (dd, *J* = 9.6, 8.1 Hz, 1 H), 4.88 (t, *J* = 9.6 Hz, 1 H), 4.48 (d, *J* = 7.8 Hz, 1 H), 3.96–3.86 (m, 1 H), 3.77 (s, 3 H, Me), 3.74 (s, 3 H, Me), 3.71–3.59 (m, 2 H), 3.59–3.50 (m, 1 H), 2.46 (td, *J* = 6.8, 2.6 Hz, 2 H), 2.30–2.18 (m, 1 H), 2.13–1.97 (m, 11 H).

^{13}C NMR (75 Hz, $CDCl_3$): δ = 170.1, 169.6, 169.3, 169.1, 169.0, 100.5, 80.4, 72.5, 71.6, 71.02, 70.95, 69.5, 67.5, 52.6, 47.5, 30.1, 20.5, 20.4, 19.6.

ESI-MS: *m*/*z* = 495 [M^+ + Na], 490 [M^+ + NH_4].

Anal. Calcd for $C_{21}H_{28}O_{12}$: C, 53.39; H, 5.97. Found: C, 53.51; H, 5.97.

Disaccharide 1k (Scheme 8)[8a]

To a three-necked flask equipped with a reflux condenser were added sequentially **4** (5.5814 g, 15.0 mmol), **5** (9.5895 g, 18.1 mmol), CH_2Cl_2 (75 mL), and $BF_3{\cdot}OEt_2$ (2.8 mL, *d* = 1.12 g/mL, 3.316 g, 22.2 mmol). The reaction was complete after the mixture had stirred at r.t. for 12 h as monitored by TLC (PE–EtOAc, 2:1). The resulting mixture was washed with H_2O (50 mL) and sat. aq $NaHCO_3$ (50 mL), and dried (Na_2SO_4). Filtration, evaporation of the solvent, and chromatography (silica gel, PE–EtOAc, 2:1 to 1.5:1) afforded 1k.

Yield: 7.9667 g (62%); solid; mp 142–144 °C (CH_2Cl_2–hexane); $[\alpha]_D^{20}$ –26.6 (*c* = 1.113, $CHCl_3$).

IR (KBr): 3280, 3063, 2959, 2941, 2881, 2120, 1736, 1599, 1452, 1371, 1316, 1281, 1257, 1227, 1177, 1094, 1069, 1037 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.96 (d, *J* = 7.8 Hz, 2 H, ArH), 7.92 (d, *J* = 7.8 Hz, 2 H, ArH), 7.82 (d, *J* = 7.5 Hz, 2 H, ArH), 7.56–7.16 (m, 9 H, ArH), 5.93 (t, *J* = 9.6 Hz, 1 H), 5.53 (t, *J* = 8.9 Hz, 1 H), 5.44 (t, *J* = 9.6 Hz, 1 H), 5.24 (t, *J* = 9.5 Hz, 1 H), 5.16–5.00 (m, 3 H), 4.69 (d, *J* = 8.1 Hz, 1 H), 4.55–4.37 (m, 2 H), 4.26 (dd, *J* = 12.5, 4.4 Hz, 1 H), 4.18–3.97 (m, 3 H), 3.90–3.66 (m, 2 H), 2.50 (s, 1 H), 2.12 (s, 3 H, Me), 2.01 (s, 9 H, Me × 3).

^{13}C NMR (75 Hz, $CDCl_3$): δ = 170.3, 169.9, 169.12, 169.09, 165.4, 165.0, 164.8, 133.4, 133.0, 129.54, 129.50, 129.4, 128.9, 128.4, 128.3, 128.2, 128.0, 100.5, 98.2, 78.1, 75.6, 73.7, 72.7, 72.4, 71.5, 71.2, 70.9, 69.4, 68.0, 61.5, 55.8, 20.5, 20.4, 20.32, 20.30.

MS (MALDI): *m*/*z* = 899 [M^+ + K], 883 [M^+ + Na].

Anal. Calcd for $C_{44}H_{44}O_{18}$: C, 61.39; H, 5.15. Found: C, 61.35; H, 5.21.

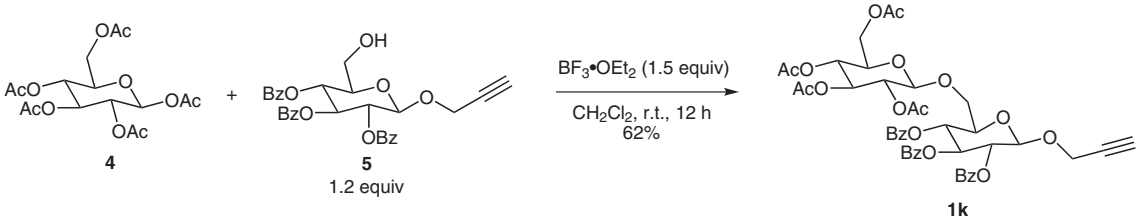
Disaccharide 1l (Scheme 9)[8a]

To a three-neck flask equipped with a reflux condenser were added sequentially **6** (3.9057 g, 10.0 mmol), **5** (6.3712 g, 12.0 mmol), CH_2Cl_2 (50 mL), and $BF_3{\cdot}OEt_2$ (1.9 mL, *d* = 1.12 g/mL, 2.128 g, 15.0 mmol). The reaction was complete after the mixture had stirred at r.t. for 12 h as monitored by TLC (PE–EtOAc, 2:1). The resulting mixture was washed with H_2O (50 mL) and sat. aq $NaHCO_3$ (50 mL), and dried (Na_2SO_4). Filtration, evaporation of the solvent, and chromatography (silica gel, PE–EtOAc, 1.5:1) afforded **1l** as a syrup.

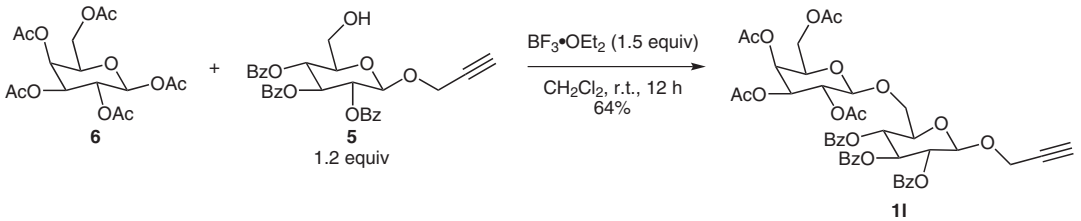
Yield: 5.5272 g (64%); $[\alpha]_D^{20}$ –21.9 (*c* = 0.92, $CHCl_3$).

IR (KBr): 3280, 3066, 2963, 2941, 2878, 2120, 1736, 1602, 1452, 1370, 1316, 1255, 1178, 1069 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.97 (d, *J* = 7.5 Hz, 2 H, ArH), 7.92 (d, *J* = 7.5 Hz, 2 H, ArH), 7.82 (d, *J* = 7.5 Hz, 2 H, ArH), 7.57–7.20 (m, 9 H, ArH), 5.91 (t, *J* = 9.6 Hz, 1 H), 5.52 (t, *J* = 8.9 Hz, 1 H), 5.46–5.36 (m, 2 H), 5.31–5.20 (m, 1 H), 5.09 (d, *J* = 7.8 Hz, 1 H), 5.03 (dd, *J* = 10.2, 3.0 Hz, 1 H), 4.62 (d, *J* = 8.1 Hz, 1 H), 4.55–4.47 (m, 2 H), 4.18–3.99 (m, 4 H), 3.92 (t, *J* = 6.5 Hz, 1 H), 3.87–



Scheme 8



Scheme 9

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3.74 (m, 1 H), 2.47 (s, 1 H), 2.13 (s, 3 H, Me), 2.12 (s, 3 H, Me), 1.99 (s, 3 H, Me), 1.97 (s, 3 H, Me).

^{13}C NMR (75 Hz, $CDCl_3$): δ = 170.2, 170.1, 170.0, 169.4, 165.6, 165.2, 165.0, 133.5, 133.1, 129.71, 129.65, 129.6, 129.1, 128.6, 128.4, 128.2, 101.2, 98.3, 78.2, 75.6, 73.8, 72.7, 71.3, 70.7, 70.6, 69.5, 68.5, 68.2, 66.8, 61.1, 55.9, 20.7, 20.5, 20.4.

MS (MALDI): *m*/*z* = 899 [M^+ + K], 883 [M^+ + Na].

Anal. Calcd for $C_{44}H_{44}O_{18}$: C, 61.39; H, 5.15. Found: C, 61.16; H, 5.30.

Carbohydrate-Based Terminal Allene 2a; Typical Procedure for Conditions A

To a round-bottom flask (25 mL) equipped with a reflux condenser were added sequentially CuI (0.1941 g, 1.0 mmol), $(HCHO)_n$ (0.7525 g, 25.0 mmol), **1a** (3.8602 g, 10.0 mmol), 1,4-dioxane (15.0 mL), and *i*-Pr_2NH (2.53 mL, *d* = 0.718 g/mL, 1.818 g, 18.0 mmol). The reaction was complete after the mixture had stirred at 110 °C for 24 h as monitored by TLC (PE–EtOAc, 2:1). After evaporation of the solvent, chromatography of the residue (silica gel, PE–EtOAc, 2.5:1 to 2:1) afforded **2a**.

Yield: 2.5569 g (64%); solid; mp 95 °C (CH_2Cl_2–hexane); $[\alpha]_D^{20}$ –9.8 (*c* = 1.00, $CHCl_3$).

IR (KBr): 2992, 2974, 2962, 2947, 2914, 2889, 1965, 1759, 1426, 1379, 1364, 1236, 1173, 1101, 1043 cm^{-1}.

^{1}H NMR (300 MHz, $CDCl_3$): δ = 5.30–4.96 (m, 4 H), 4.87–4.77 (m, 2 H), 4.62 (d, *J* = 7.8 Hz, 1 H), 4.38–4.22 (m, 2 H), 4.22–4.09 (m, 2 H), 3.74–3.64 (m, 1 H), 2.09 (s, 3 H, Me), 2.05 (s, 3 H, Me), 2.03 (s, 3 H, Me), 2.01 (s, 3 H, Me).

^{13}C NMR (75 Hz, $CDCl_3$): δ = 209.4, 170.4, 170.0, 169.2, 169.1, 98.9, 86.6, 75.9, 72.6, 71.6, 71.0, 68.2, 66.8, 61.7, 20.51, 20.47, 20.4.

ESI-MS: *m*/*z* = 423 [M^+ + Na], 418 [M^+ + NH_4].

Anal. Calcd for $C_{18}H_{24}O_{10}$: C, 54.00; H, 6.04. Found: C, 54.06; H, 6.01.

Carbohydrate-Based Terminal Allene 2a; Typical Procedure for Conditions B

To a round-bottom flask (25 mL) equipped with a reflux condenser were added sequentially CuI (0.9552 g, 5.0 mmol), $(HCHO)_n$ (0.7506 g, 25.0 mmol), **1a** (3.8629 g, 10.0 mmol), 1,4-dioxane (15.0 mL), and Cy_2NH (3.58 mL, *d* = 0.91 g/mL, 3.258 g, 18.0 mmol). The reaction was complete after the mixture had stirred at 110 °C for 12 h as monitored by TLC (PE–EtOAc, 2:1). After evaporation of the solvent, chromatography (silica gel, PE–EtOAc, 2.5:1 to 2:1) afforded **2a**.

Yield: 2.8925 g (72%).

^{1}H NMR (300 MHz, $CDCl_3$): δ = 5.29–4.96 (m, 4 H), 4.87–4.77 (m, 2 H), 4.62 (d, *J* = 7.8 Hz, 1 H), 4.38–4.22 (m, 2 H), 4.22–4.09 (m, 2 H), 3.76–3.66 (m, 1 H), 2.09 (s, 3 H, Me), 2.05 (s, 3 H, Me), 2.03 (s, 3 H, Me), 2.01 (s, 3 H, Me).

Carbohydrate-Based Terminal Allene 2b

Conditions A: According to the Typical Procedure for Conditions A, the reaction of CuI (0.1920 g, 1.0 mmol), $(HCHO)_n$ (0.7511 g, 25.0 mmol), **1b** (4.9837 g, 10.0 mmol), and *i*-Pr_2NH (2.53 mL, *d* = 0.718 g/mL, 1.818 g, 18.0 mmol) in 1,4-dioxane (15.0 mL) afforded **2b** after chromatography (silica gel, PE–EtOAc, 2:1).

Yield: 2.6465 g (52%); solid; mp 110–111 °C (CH_2Cl_2–hexane); $[\alpha]_D^{20}$ +9.7 (*c* = 1.13, $CHCl_3$).

IR (KBr): 2961, 2872, 2810, 1960, 1752, 1598, 1442, 1376, 1354, 1310, 1236, 1224, 1192, 1173, 1116, 1097, 1040 cm^{-1}.

^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.78 (d, *J* = 8.1 Hz, 2 H, ArH), 7.36 (d, *J* = 7.8 Hz, 2 H, ArH), 5.23–5.08 (m, 2 H), 4.97–4.77 (m, 4 H), 4.56 (d, *J* = 8.1 Hz, 1 H), 4.30–4.18 (m, 1 H), 4.18–4.00 (m, 3 H), 3.81–3.69 (m, 1 H), 2.46 (s, 3 H, Me), 2.03 (s, 3 H, Me), 2.01 (s, 3 H, Me), 1.99 (s, 3 H, Me).

^{13}C NMR (75 Hz, $CDCl_3$): δ = 209.4, 170.1, 169.4, 169.2, 145.1, 132.2, 129.8, 128.0, 99.0, 86.7, 76.1, 72.4, 71.5, 70.9, 68.5, 67.6, 66.9, 21.6, 20.54, 20.46, 20.4.

ESI-MS: *m*/*z* = 551 [M^+ + K], 535 [M^+ + Na], 530 [M^+ + NH_4].

Anal. Calcd for $C_{23}H_{28}O_{11}S$: C, 53.90; H, 5.51. Found: C, 53.91; H, 5.47.

Conditions B: According to the Typical Procedure for Conditions B, the reaction of CuI (0.9530 g, 5.0 mmol), $(HCHO)_n$ (0.7481 g, 25.0 mmol), **1b** (4.9812 g, 10.0 mmol), and Cy_2NH (3.58 mL, *d* = 0.91 g/mL, 3.258 g, 18.0 mmol) in 1,4-dioxane (15.0 mL) afforded **2b** after chromatography (silica gel, PE–EtOAc, 2:1).

Yield: 2.7449 g (54%).

^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.78 (d, *J* = 8.4 Hz, 2 H, ArH), 7.36 (d, *J* = 8.1 Hz, 2 H, ArH), 5.23–5.08 (m, 2 H), 4.96–4.76 (m, 4 H), 4.56 (d, *J* = 7.8 Hz, 1 H), 4.29–4.18 (m, 1 H), 4.17–4.00 (m, 3 H), 3.81–3.69 (m, 1 H), 2.46 (s, 3 H, Me), 2.03 (s, 3 H, Me), 2.01 (s, 3 H, Me), 1.99 (s, 3 H, Me).

Carbohydrate-Based Terminal Allene 2c

Conditions A: According to the Typical Procedure for Conditions A, the reaction of CuI (0.0958 g, 0.5 mmol), $(HCHO)_n$ (0.3775 g, 12.5 mmol), a solution of **1c** (*w*/*w* = 41.61% in 1,4-dioxane) (5.6390 g, 5.0 mmol), *i*-Pr_2NH (1.27 mL, *d* = 0.718 g/mL, 0.912 g, 9.0 mmol), and 1,4-dioxane (3.5 mL) afforded **2c** after chromatography (silica gel, PE–EtOAc, 1:1).

Yield: 1.5526 g (64%).

^{1}H NMR (300 MHz, $CDCl_3$): δ = 5.88 (t, *J* = 5.6 Hz, 1 H, NH), 5.25–5.12 (m, 2 H), 5.03–4.79 (m, 4 H), 4.60 (d, *J* = 7.8 Hz, 1 H), 4.36–4.25 (m, 1 H), 4.23–4.11 (m, 1 H), 3.65–3.51 (m, 2 H), 3.50–3.37 (m, 1 H), 2.23–2.12 (m, 2 H, CH_2), 2.06 (s, 3 H, Me), 2.05 (s, 3 H, Me), 2.00 (s, 3 H, Me), 1.68–1.53 (m, 2 H, CH_2), 1.39–1.20 (m, 8 H, CH_2 × 4), 0.88 (t, *J* = 6.6 Hz, 3 H, Me).

Conditions B: According to the Typical Procedure for Conditions B, the reaction of CuI (0.4769 g, 2.5 mmol), $(HCHO)_n$ (0.3754 g, 12.5 mmol), a solution of **1c** (*w*/*w* = 41.61% in 1,4-dioxane) (5.6628 g, 5.0 mmol), Cy_2NH (1.79 mL, *d* = 0.91 g/mL, 1.629 g, 9.0 mmol) and 1,4-dioxane (3.5 mL) afforded **2c** after chromatography (silica gel, PE–EtOAc, 1:1).

Yield: 1.3260 g (55%); syrup; $[\alpha]_D^{20}$ –4.9 (*c* = 1.03, $CHCl_3$).

IR (neat): 3316, 2931, 2859, 1956, 1759, 1654, 1535, 1438, 1368, 1245, 1220, 1165, 1052 cm^{-1}.

MS (ESI): *m*/*z* = 506 [M^+ + Na], 484 [M^+ + H].

^{1}H NMR (300 MHz, $CDCl_3$): δ = 5.85 (t, *J* = 5.6 Hz, 1 H, NH), 5.28–5.13 (m, 2 H), 5.04–4.77 (m, 4 H), 4.60 (d, *J* = 8.1 Hz, 1 H), 4.37–4.25 (m, 1 H), 4.23–4.10 (m, 1 H), 3.66–3.51 (m, 2 H), 3.50–3.37 (m, 1 H), 2.23–2.12 (m, 2 H, CH_2), 2.06 (s, 3 H, Me), 2.05 (s, 3 H, Me), 2.00 (s, 3 H, Me), 1.68–1.53 (m, 2 H, CH_2), 1.40–1.20 (m, 8 H, CH_2 × 4), 0.88 (t, *J* = 6.5 Hz, 3 H, Me).

^{13}C NMR (75 Hz, $CDCl_3$): δ = 209.5, 173.2, 170.2, 169.6, 169.3, 99.2, 86.7, 76.1, 72.6, 72.5, 71.2, 68.9, 67.2, 38.9, 36.6, 31.6, 29.2, 28.9, 25.5, 22.5, 20.6, 20.5, 14.0.

HRMS: *m*/*z* calcd for $C_{24}H_{38}NO_9$ [M^+ + H]: 484.2541; found: 484.2549.

Carbohydrate-Based Terminal Allene 2d

Conditions A: According to the Typical Procedure for Conditions A, the reaction of CuI (0.0574 g, 0.3 mmol), $(HCHO)_n$ (0.2257 g, 7.5 mmol), **1d** (1.3750 g, 3.0 mmol), and *i*-Pr_2NH (0.76 mL, *d* = 0.718 g/mL, 0.5454 g, 5.4 mmol) in 1,4-dioxane (4.5 mL) afforded **2d** after chromatography (silica gel, PE–EtOAc, 2:1).

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Yield: 0.7101 g (50%); solid; mp 82–83 °C (CH_2Cl_2–hexane); $[\alpha]_D^{20}$ +9.8 (*c* = 0.96, $CHCl_3$).

IR (KBr): 2959, 1956, 1755, 1436, 1369, 1245, 1221, 1156, 1044 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.24–5.11 (m, 2 H), 5.00–4.77 (m, 4 H), 4.52 (d, *J* = 8.1 Hz, 1 H), 4.33–4.23 (m, 1 H), 4.18–4.07 (m, 1 H), 3.76 (s, 3 H, Me), 3.74 (s, 3 H, Me), 3.72–3.64 (m, 1 H), 3.53 (td, *J* = 9.9, 2.7 Hz, 1 H), 2.30–2.18 (m, 1 H), 2.13–1.97 (m, 1 H), 2.06 (s, 3 H, Me), 2.04 (s, 3 H, Me), 2.00 (s, 3 H, Me).

^{13}C NMR (75 Hz, $CDCl_3$): δ = 209.4, 170.1, 169.6, 169.2, 169.12, 169.09, 99.0, 86.7, 76.0, 72.6, 71.7, 71.2, 71.0, 66.8, 52.61, 52.58, 47.5, 30.3, 20.54, 20.46.

ESI-MS: *m*/*z* = 495 [M^+ + Na].

Anal. Calcd for $C_{21}H_{28}O_{12}$: C, 53.39; H, 5.97. Found: C, 53.47; H, 5.90.

Conditions B: According to the Typical Procedure for Conditions B, the reaction of CuI (0.2867 g, 1.5 mmol), $(HCHO)_n$ (0.2273 g, 7.5 mmol), **1d** (1.3746 g, 3.0 mmol), and Cy_2NH (1.07 mL, *d* = 0.91 g/mL, 0.9737 g, 5.4 mmol) in 1,4-dioxane (4.5 mL) afforded **2d** after chromatography (silica gel, PE–EtOAc, 2:1).

Yield: 0.8410 g (59%).

^1H NMR (300 MHz, $CDCl_3$): δ = 5.25–5.12 (m, 2 H), 5.01–4.78 (m, 4 H), 4.52 (d, *J* = 8.1 Hz, 1 H), 4.33–4.23 (m, 1 H), 4.18–4.07 (m, 1 H), 3.76 (s, 3 H, Me), 3.74 (s, 3 H, Me), 3.71–3.65 (m, 1 H), 3.57 (td, *J* = 9.6, 2.4 Hz, 1 H), 2.30–2.18 (m, 1 H), 2.14–1.96 (m, 1 H), 2.06 (s, 3 H, Me), 2.04 (s, 3 H, Me), 2.00 (s, 3 H, Me).

Carbohydrate-Based Terminal Allene 2e

Conditions A: According to the Typical Procedure for Conditions A, the reaction of CuI (0.1937 g, 1.0 mmol), $(HCHO)_n$ (0.7540 g, 25.0 mmol), **1e** (4.0391 g, 10.0 mmol), and *i*-Pr_2NH (2.53 mL, *d* = 0.718 g/mL, 1.817 g, 18.0 mmol) in 1,4-dioxane (15.0 mL) afforded **2e** after chromatography (silica gel, PE–EtOAc, 2:1).

Yield: 2.1454 g (51%); solid; mp 70–71 °C (CH_2Cl_2–hexane); $[\alpha]_D^{20}$ –14.9 (*c* = 1.03, $CHCl_3$).

IR (KBr): 2944, 2885, 1957, 1755, 1434, 1368, 1228, 1170, 1039 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.21 (t, *J* = 9.5 Hz, 1 H), 5.14–4.95 (m, 3 H), 4.74–4.65 (m, 2 H), 4.53 (d, *J* = 7.8 Hz, 1 H), 4.28 (dd, *J* = 12.3, 4.5 Hz, 1 H), 4.14 (dd, *J* = 12.3, 2.1 Hz, 1 H), 4.00–3.88 (m, 1 H), 3.76–3.65 (m, 1 H), 3.63–3.50 (m, 1 H), 2.36–2.22 (m, 2 H), 2.09 (s, 3 H, Me), 2.05 (s, 3 H, Me), 2.03 (s, 3 H, Me), 2.01 (s, 3 H, Me).

^{13}C NMR (75 Hz, $CDCl_3$): δ = 208.7, 170.5, 170.2, 169.3, 169.2, 100.7, 86.0, 75.1, 72.6, 71.6, 71.1, 69.2, 68.2, 61.8, 28.4, 20.6, 20.49, 20.46.

ESI-MS: *m*/*z* = 437 [M^+ + Na].

Anal. Calcd for $C_{19}H_{26}O_{10}$: C, 55.07; H, 6.32. Found: C, 55.45; H, 6.50.

Conditions B: According to the Typical Procedure for Conditions B, the reaction of CuI (0.9572 g, 5.0 mmol), $(HCHO)_n$ (0.7508 g, 25.0 mmol), **1e** (4.0053 g, 10.0 mmol), and Cy_2NH (3.58 mL, *d* = 0.91 g/mL, 3.2580 g, 18.0 mmol) in 1,4-dioxane (15.0 mL) afforded **2e** after chromatography (silica gel, PE–EtOAc, 2:1).

Yield: 2.7283 g (66%).

^1H NMR (300 MHz, $CDCl_3$): δ = 5.21 (t, *J* = 9.5 Hz, 1 H), 5.13–4.95 (m, 3 H), 4.73–4.65 (m, 2 H), 4.53 (d, *J* = 7.8 Hz, 1 H), 4.28 (dd, *J* = 12.3, 4.8 Hz, 1 H), 4.14 (dd, *J* = 12.3, 2.4 Hz, 1 H), 3.98–3.88 (m, 1 H), 3.74–3.66 (m, 1 H), 3.62–3.52 (m, 1 H), 2.35–2.22 (m, 2 H), 2.09 (s, 3 H, Me), 2.05 (s, 3 H, Me), 2.03 (s, 3 H, Me), 2.01 (s, 3 H, Me).

Carbohydrate-Based Terminal Allene 2f

Conditions A: According to the Typical Procedure for Conditions A, the reaction of CuI (0.0961 g, 0.5 mmol), $(HCHO)_n$ (0.3745 g, 12.5 mmol), **1f** (2.5541 g, 5.0 mmol), and *i*-Pr_2NH (1.27 mL, *d* = 0.718 g/mL, 0.909 g, 9.0 mmol) in 1,4-dioxane (7.5 mL) afforded **2f** after chromatography (silica gel, PE–EtOAc, 2:1).

Yield: 1.3613 g (52%); solid; mp 86–87 °C (CH_2Cl_2–hexane); $[\alpha]_D^{20}$ –2.6 (*c* = 1.005, $CHCl_3$).

IR (KBr): 3033, 2982, 2958, 2892, 1957, 1754, 1598, 1428, 1358, 1218, 1173, 1097, 1051 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 7.78 (d, *J* = 8.1 Hz, 2 H, ArH), 7.36 (d, *J* = 8.1 Hz, 2 H, ArH), 5.17 (t, *J* = 9.5 Hz, 1 H), 5.12–5.00 (m, 1 H), 4.96–4.83 (m, 2 H), 4.75–4.63 (m, 2 H), 4.46 (d, *J* = 7.8 Hz, 1 H), 4.16–4.00 (m, 2 H), 3.90–3.59 (m, 2 H), 3.57–3.45 (m, 1 H), 2.46 (s, 3 H, Me), 2.31–2.18 (m, 2 H), 2.03 (s, 3 H, Me), 2.01 (s, 3 H, Me), 1.99 (s, 3 H, Me).

^{13}C NMR (75 Hz, $CDCl_3$): δ = 208.7, 170.0, 169.3, 169.1, 145.0, 132.2, 129.7, 127.9, 100.4, 85.9, 75.1, 72.3, 71.3, 70.8, 69.0, 68.5, 67.6, 28.3, 21.5, 20.42, 20.40, 20.35.

ESI-MS: *m*/*z* = 565 [M^+ + K], 549 [M^+ + Na], 544 [M^+ + NH_4].

Anal. Calcd for $C_{24}H_{30}O_{11}S$: C, 54.74; H, 5.74. Found: C, 54.91; H, 5.80.

Conditions B: According to the Typical Procedure for Conditions B, the reaction of CuI (0.4763 g, 2.5 mmol), $(HCHO)_n$ (0.3753 g, 12.5 mmol), **1f** (2.5597 g, 5.0 mmol), and Cy_2NH (1.80 mL, *d* = 0.91 g/mL, 1.638 g, 9.0 mmol) in 1,4-dioxane (7.5 mL) afforded **2f** after chromatography (silica gel, PE–EtOAc, 2:1).

Yield: 1.4281 g (54%).

^1H NMR (300 MHz, $CDCl_3$): δ = 7.78 (d, *J* = 8.4 Hz, 2 H, ArH), 7.36 (d, *J* = 8.4 Hz, 2 H, ArH), 5.17 (t, *J* = 9.6 Hz, 1 H), 5.12–5.00 (m, 1 H), 4.96–4.85 (m, 2 H), 4.75–4.65 (m, 2 H), 4.46 (d, *J* = 7.8 Hz, 1 H), 4.16–4.00 (m, 2 H), 3.89–3.60 (m, 2 H), 3.57–3.45 (m, 1 H), 2.46 (s, 3 H, Me), 2.31–2.19 (m, 2 H), 2.03 (s, 3 H, Me), 2.00 (s, 3 H, Me), 1.99 (s, 3 H, Me).

Carbohydrate-Based Terminal Allene 2g

Conditions A: According to the Typical Procedure for Conditions A, the reaction of CuI (0.0575 g, 0.3 mmol), $(HCHO)_n$ (0.2251 g, 7.5 mmol), **1g** (1.4619 g, 3.0 mmol), and *i*-Pr_2NH (0.76 mL, *d* = 0.718 g/mL, 0.5454 g, 5.4 mmol) in 1,4-dioxane (4.5 mL) afforded **2g** after chromatography (silica gel, PE–EtOAc, 2:1).

Yield: 0.8905 g (59%).

^1H NMR (300 MHz, $CDCl_3$): δ = 5.22–5.02 (m, 2 H), 4.99–4.83 (m, 2 H), 4.74–4.65 (m, 2 H), 4.43 (d, *J* = 8.1 Hz, 1 H), 3.94–3.84 (m, 1 H), 3.76 (s, 3 H, Me), 3.74 (s, 3 H, Me), 3.68 (dd, *J* = 9.6, 5.4 Hz, 1 H), 3.59–3.46 (m, 2 H), 2.32–2.17 (m, 3 H), 2.15–1.95 (m, 1 H), 2.06 (s, 3 H, Me), 2.03 (s, 3 H, Me), 2.00 (s, 3 H, Me).

Conditions B: According to the Typical Procedure for Conditions B, the reaction of CuI (0.2891 g, 1.5 mmol), $(HCHO)_n$ (0.2270 g, 7.5 mmol), **1g** (1.4629 g, 3.0 mmol), and Cy_2NH (1.07 mL, *d* = 0.91 g/mL, 0.9774 g, 5.4 mmol) in 1,4-dioxane (4.5 mL) afforded **2g** after chromatography (silica gel, PE–EtOAc, 2:1).

Yield: 1.0100 g (67%); solid; mp 84–85 °C (CH_2Cl_2–hexane); $[\alpha]_D^{20}$ +2.3 (*c* = 0.98, $CHCl_3$).

IR (KBr): 2960, 2899, 2852, 1956, 1755, 1724, 1435, 1369, 1244, 1216, 1157, 1039, 1003 cm^{-1}.

^1H NMR (300 MHz, $CDCl_3$): δ = 5.22–5.03 (m, 2 H), 5.00–4.83 (m, 2 H), 4.74–4.66 (m, 2 H), 4.42 (d, *J* = 7.8 Hz, 1 H), 3.95–3.84 (m, 1 H), 3.76 (s, 3 H, Me), 3.74 (s, 3 H, Me), 3.68 (dd, *J* = 9.8, 5.3 Hz, 1 H), 3.58–3.47 (m, 2 H), 2.32–2.18 (m, 3 H), 2.12–1.96 (m, 1 H), 2.06 (s, 3 H, Me), 2.04 (s, 3 H, Me), 2.00 (s, 3 H, Me).

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^{13}C NMR (75 Hz, $CDCl_3$): δ = 208.7, 170.1, 169.5, 169.2, 169.1, 100.6, 86.0, 75.0, 72.6, 71.8, 71.2, 70.9, 69.0, 52.6, 52.5, 47.5, 30.2, 28.4, 20.53, 20.48, 20.45.

ESI-MS: *m*/*z* = 509 [M^+ + Na], 504 [M^+ + NH_4].

Anal. Calcd for $C_{22}H_{30}O_{12}$: C, 54.32; H, 6.22. Found: C, 54.53; H, 6.15.

Carbohydrate-Based Terminal Allene 2h

Conditions A: According to the Typical Procedure for Conditions A, the reaction of CuI (0.1920 g, 1.0 mmol), $(HCHO)_n$ (0.7513 g, 25.0 mmol), **1h** (4.0102 g, 10.0 mmol), and *i*-Pr_2NH (2.53 mL, *d* = 0.718 g/mL, 1.818 g, 18.0 mmol) in 1,4-dioxane (15.0 mL) afforded **2h** as a syrup after chromatography (silica gel, PE–EtOAc, 2.5:1).

Yield: 2.9465 g (71%); $[\alpha]_D^{20}$ +23.9 (*c* = 1.253, $CHCl_3$).

IR (neat): 2952, 2869, 1950, 1755, 1435, 1372, 1227, 1039 cm^{-1}.

^{1}H NMR (300 MHz, $CDCl_3$): δ = 5.31–5.00 (m, 4 H), 4.90–4.78 (m, 2 H), 4.59 (d, *J* = 9.9 Hz, 1 H), 4.25 (dd, *J* = 12.5, 5.0 Hz, 1 H), 4.14 (dd, *J* = 12.3, 2.1 Hz, 1 H), 3.73–3.63 (m, 1 H), 3.48–3.35 (m, 1 H), 3.28–3.15 (m, 1 H), 2.08 (s, 3 H, Me), 2.06 (s, 3 H, Me), 2.04 (s, 3 H, Me), 2.02 (s, 3 H, Me).

^{13}C NMR (75 Hz, $CDCl_3$): δ = 209.4, 170.5, 170.0, 169.2, 87.0, 82.0, 76.1, 75.7, 73.7, 69.7, 68.2, 61.9, 28.9, 20.6, 20.5, 20.4.

ESI-MS: *m*/*z* = 455 [M^+ + K], 439 [M^+ + Na], 434 [M^+ + NH_4].

HRMS: *m*/*z* calcd for $C_{18}H_{24}NaO_9S$ [M^+ + Na]: 439.1033; found: 439.1032.

Conditions B: According to the Typical Procedure for Conditions B, the reaction of CuI (1.1475 g, 6.0 mmol), $(HCHO)_n$ (0.8994 g, 30.0 mmol), **1h** (4.8015 g, 11.9 mmol), and Cy_2NH (4.30 mL, *d* = 0.91 g/mL, 3.9096 g, 21.6 mmol) in 1,4-dioxane (18.0 mL) afforded **2h** after chromatography (silica gel, PE–EtOAc, 2.5:1).

Yield: 3.5093 g (71%).

^{1}H NMR (300 MHz, $CDCl_3$): δ = 5.30–5.00 (m, 4 H), 4.88–4.79 (m, 2 H), 4.59 (d, *J* = 10.2 Hz, 1 H), 4.25 (dd, *J* = 12.3, 5.1 Hz, 1 H), 4.14 (dd, *J* = 12.3, 2.1 Hz, 1 H), 3.72–3.63 (m, 1 H), 3.46–3.35 (m, 1 H), 3.27–3.15 (m, 1 H), 2.08 (s, 3 H, Me), 2.06 (s, 3 H, Me), 2.04 (s, 3 H, Me), 2.02 (s, 3 H, Me).

Carbohydrate-Based Terminal Allene 2i

Conditions A: According to the Typical Procedure for Conditions A, the reaction of CuI (0.1908 g, 1.0 mmol), $(HCHO)_n$ (0.7559 g, 25.0 mmol), **1i** (3.8614 g, 10.0 mmol), and *i*-Pr_2NH (2.53 mL, *d* = 0.718 g/mL, 1.818 g, 18.0 mmol) in 1,4-dioxane (15.0 mL) afforded **2i** after chromatography (silica gel, PE–EtOAc, 2:1).

Yield: 2.3305 g (58%); solid; mp 103–104 °C (EtOAc–hexane); $[\alpha]_D^{20}$ +3.8 (*c* = 1.053, $CHCl_3$).

IR (KBr): 2979, 2943, 2852, 2804, 1952, 1748, 1370, 1225, 1163, 1124, 1084, 1046 cm^{-1}.

^{1}H NMR (300 MHz, $CDCl_3$): δ = 5.39 (d, *J* = 2.7 Hz, 1 H), 5.26–5.15 (m, 2 H), 5.04 (dd, *J* = 10.4, 3.0 Hz, 1 H), 4.86–4.77 (m, 2 H), 4.61 (d, *J* = 8.1 Hz, 1 H), 4.38–4.27 (m, 1 H), 4.23–4.10 (m, 3 H), 3.95 (t, *J* = 6.5 Hz, 1 H), 2.16 (s, 3 H, Me), 2.06 (s, 3 H, Me), 2.05 (s, 3 H, Me), 1.98 (s, 3 H, Me).

^{13}C NMR (75 Hz, $CDCl_3$): δ = 209.1, 169.93, 169.85, 169.7, 169.0, 99.3, 86.6, 75.7, 70.6, 70.3, 68.4, 66.7, 66.6, 60.9, 20.4, 20.3, 20.2.

ESI-MS: *m*/*z* = 418 [M^+ + NH_4].

Anal. Calcd for $C_{18}H_{24}O_{10}$: C, 54.00; H, 6.04. Found: C, 54.22; H, 6.02.

Conditions B: According to the Typical Procedure for Conditions B, the reaction of CuI (0.9613 g, 5.0 mmol), $(HCHO)_n$ (0.7509 g, 25.0 mmol), **1i** (3.8609 g, 10.0 mmol), and Cy_2NH (3.58 mL, *d* = 0.91 g/mL, 3.258 g, 18.0 mmol) in 1,4-dioxane (15.0 mL) afforded **2i** after chromatography (silica gel, PE–EtOAc, 2:1).

Yield: 2.7709 g (69%).

^{1}H NMR (300 MHz, $CDCl_3$): δ = 5.40 (d, *J* = 3.0 Hz, 1 H), 5.27–5.13 (m, 2 H), 5.03 (dd, *J* = 10.5, 3.3 Hz, 1 H), 4.86–4.77 (m, 2 H), 4.58 (d, *J* = 8.1 Hz, 1 H), 4.39–4.28 (m, 1 H), 4.24–4.08 (m, 3 H), 3.92 (t, *J* = 6.6 Hz, 1 H), 2.16 (s, 3 H, Me), 2.07 (s, 3 H, Me), 2.06 (s, 3 H, Me), 1.99 (s, 3 H, Me).

Carbohydrate-Based Terminal Allene 2j

Conditions A: According to the Typical Procedure for Conditions A, the reaction of CuI (0.1398 g, 0.73 mmol), $(HCHO)_n$ (0.5467 g, 18.18 mmol), **1j** (2.9241 g, 7.27 mmol), and *i*-Pr_2NH (1.84 mL, *d* = 0.718 g/mL, 1.3217 g, 13.09 mmol) in 1,4-dioxane (10.9 mL) afforded **2j** as a syrup after chromatography (silica gel, PE–EtOAc, 3:1):

Yield: 2.0270 g, 67%; $[\alpha]_D^{20}$ +26.9 (*c* = 0.72, $CHCl_3$).

IR (neat): 2970, 2938, 2863, 1950, 1746, 1433, 1371, 1225, 1151, 1083, 1054 cm^{-1}.

^{1}H NMR (300 MHz, $CDCl_3$): δ = 5.44 (d, *J* = 3.0 Hz, 1 H), 5.32–5.13 (m, 2 H), 5.07 (dd, *J* = 10.2, 3.2 Hz, 1 H), 4.89–4.78 (m, 2 H), 4.58 (d, *J* = 10.1 Hz, 1 H), 4.22–4.06 (m, 2 H), 3.92 (t, *J* = 6.6 Hz, 1 H), 3.49–3.35 (m, 1 H), 3.30–3.18 (m, 1 H), 2.17 (s, 3 H, Me), 2.08 (s, 3 H, Me), 2.05 (s, 3 H, Me), 1.99 (s, 3 H, Me).

^{13}C NMR (75 Hz, $CDCl_3$): δ = 209.4, 170.2, 170.1, 169.9, 169.4, 87.0, 82.5, 76.0, 74.2, 71.7, 67.1, 67.0, 61.3, 28.9, 20.6, 20.5, 20.4.

ESI-MS: *m*/*z* = 434 [M^+ + NH_4], 417 [M^+ + H].

HRMS: *m*/*z* calcd for $C_{18}H_{24}NaO_9S$ [M^+ + Na]: 439.1033; found: 439.1048.

Conditions B: According to the Typical Procedure for Conditions B, the reaction of CuI (0.4845 g, 2.54 mmol), $(HCHO)_n$ (0.3821 g, 12.7 mmol), **1j** (2.0410 g, 5.08 mmol), and Cy_2NH (1.82 mL, *d* = 0.91 g/mL, 1.6562 g, 9.14 mmol) in 1,4-dioxane (7.6 mL) afforded **2j** after chromatography (silica gel, PE–EtOAc, 3:1).

Yield: 1.4479 g (69%).

^{1}H NMR (300 MHz, $CDCl_3$): δ = 5.44 (d, *J* = 3.0 Hz, 1 H), 5.31–5.14 (m, 2 H), 5.07 (dd, *J* = 9.9, 3.3 Hz, 1 H), 4.87–4.80 (m, 2 H), 4.58 (d, *J* = 9.9 Hz, 1 H), 4.22–4.07 (m, 2 H), 3.91 (t, *J* = 6.5 Hz, 1 H), 3.48–3.36 (m, 1 H), 3.30–3.17 (m, 1 H), 2.17 (s, 3 H, Me), 2.08 (s, 3 H, Me), 2.05 (s, 3 H, Me), 1.99 (s, 3 H, Me).

Disaccharide-Based Terminal Allene 2k

Conditions A: According to the Typical Procedure for Conditions A, the reaction of CuI (0.0189 g, 0.1 mmol), $(HCHO)_n$ (0.0761 g, 2.5 mmol), **1k** (0.8609 g, 1.0 mmol), and *i*-Pr_2NH (0.25 mL, *d* = 0.718 g/mL, 0.1795 g, 1.8 mmol) in 1,4-dioxane (3.0 mL) afforded **2k** after chromatography (silica gel, PE–EtOAc, 1.5:1).

Yield: 0.6106 g (70%).

^{1}H NMR (300 MHz, $CDCl_3$): δ = 8.00–7.86 (m, 4 H, ArH), 7.81 (d, *J* = 7.5 Hz, 2 H, ArH), 7.56–7.19 (m, 9 H, ArH), 5.91 (t, *J* = 9.6 Hz, 1 H), 5.52 (t, *J* = 8.7 Hz, 1 H), 5.42 (t, *J* = 9.6 Hz, 1 H), 5.27–4.98 (m, 4 H), 4.93 (d, *J* = 8.1 Hz, 1 H), 4.80–4.71 (m, 2 H), 4.67 (d, *J* = 7.8 Hz, 1 H), 4.45–4.34 (m, 1 H), 4.30–4.18 (m, 2 H), 4.11–3.96 (m, 3 H), 3.83 (dd, *J* = 11.1, 8.1 Hz, 1 H), 3.76–3.66 (m, 1 H), 2.09 (s, 3 H, Me), 2.01 (s, 3 H, Me), 2.00 (s, 3 H, Me), 1.99 (s, 3 H, Me).

Conditions B: According to the Typical Procedure for Conditions B, the reaction of CuI (0.0956 g, 0.5 mmol), $(HCHO)_n$ (0.0764 g, 2.5 mmol), **1k** (0.8601 g, 1.0 mmol), and Cy_2NH (0.36 mL, *d* = 0.91 g/mL, 0.3276 g, 1.8 mmol) in 1,4-dioxane (3.0 mL) afforded **2k** after chromatography (silica gel, PE–EtOAc, 1.5:1).

Yield: 0.6242 g (71%); solid; mp 157–158 °C (CH_2Cl_2–hexane); $[\alpha]_D^{20}$ + 2.9 (*c* = 1.01, $CHCl_3$).

IR (KBr): 3069, 2964, 2941, 2887, 1953, 1756, 1733, 1453, 1370, 1290, 1256, 1221, 1101, 1070, 1034 cm^{-1}.

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^{1}H NMR (300 MHz, $CDCl_3$): δ = 8.02–7.87 (m, 4 H, ArH), 7.81 (d, *J* = 7.5 Hz, 2 H, ArH), 7.57–7.20 (m, 9 H, ArH), 5.89 (t, *J* = 9.6 Hz, 1 H), 5.50 (t, *J* = 8.9 Hz, 1 H), 5.40 (t, *J* = 9.6 Hz, 1 H), 5.26–4.96 (m, 4 H), 4.91 (d, *J* = 7.8 Hz, 1 H), 4.81–4.70 (m, 2 H), 4.66 (d, *J* = 7.8 Hz, 1 H), 4.44–4.34 (m, 1 H), 4.30–4.16 (m, 2 H), 4.11–3.95 (m, 3 H), 3.81 (dd, *J* = 11.4, 8.1 Hz, 1 H), 3.74–3.64 (m, 1 H), 2.09 (s, 3 H, Me), 2.01 (s, 3 H, Me), 2.003 (s, 3 H, Me), 1.995 (s, 3 H, Me).

^{13}C NMR (75 Hz, $CDCl_3$): δ = 209.3, 170.4, 170.0, 169.2, 169.1, 165.6, 165.2, 164.9, 133.5, 133.1, 129.6, 129.5, 129.1, 128.6, 128.4, 128.3, 128.2, 128.1, 100.6, 99.3, 86.7, 76.1, 73.7, 72.8, 72.6, 71.6, 71.5, 71.0, 69.5, 68.3, 68.1, 66.8, 61.6, 20.6, 20.43, 20.42, 20.39.

MS (MALDI): *m*/*z* = 913 [M^+ + K], 897 [M^+ + Na].

Anal. Calcd for $C_{45}H_{46}O_{18}$: C, 61.78; H, 5.30. Found: C, 61.42; H, 5.33.

Disaccharide-Based Terminal Allene 2l

Conditions A: According to the Typical Procedure for Conditions A, the reaction of CuI (0.0193 g, 0.1 mmol), $(HCHO)_n$ (0.0754 g, 2.5 mmol), **1l** (0.8605 g, 1.0 mmol), and *i*-Pr_2NH (0.25 mL, *d* = 0.718 g/mL, 0.1795 g, 1.8 mmol) in 1,4-dioxane (3.0 mL) afforded **2l** after chromatography (silica gel, PE–EtOAc, 1.5:1).

Yield: 0.6547 g (75%); solid; mp 87–89 °C (CH_2Cl_2–hexane); $[\alpha]_D^{20}$ + 5.0 (*c* = 1.063, $CHCl_3$).

IR (KBr): 3072, 2944, 2881, 1956, 1751, 1735, 1602, 1452, 1370, 1281, 1254, 1224, 1178, 1093, 1070, 1027 cm^{-1}.

^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.96 (d, *J* = 7.8 Hz, 2 H, ArH), 7.92 (d, *J* = 7.5 Hz, 2 H, ArH), 7.81 (d, *J* = 7.2 Hz, 2 H, ArH), 7.57–7.20 (m, 9 H, ArH), 5.89 (t, *J* = 9.8 Hz, 1 H), 5.51 (t, *J* = 8.9 Hz, 1 H), 5.45–5.34 (m, 2 H), 5.30–5.20 (m, 1 H), 5.20–5.10 (m, 1 H), 5.03 (dd, *J* = 10.5, 3.3 Hz, 1 H), 4.92 (d, *J* = 7.8 Hz, 1 H), 4.81–4.69 (m, 2 H), 4.62 (d, *J* = 7.8 Hz, 1 H), 4.46–4.35 (m, 1 H), 4.29–4.17 (m, 1 H), 4.15–3.99 (m, 4 H), 3.92 (t, *J* = 6.5 Hz, 1 H), 3.80 (dd, *J* = 11.3, 8.3 Hz, 1 H), 2.111 (s, 3 H, Me), 2.105 (s, 3 H, Me), 1.98 (s, 3 H, Me), 1.95 (s, 3 H, Me).

^{13}C NMR (75 Hz, $CDCl_3$): δ = 209.3, 170.1, 170.0, 169.9, 169.2, 165.5, 165.1, 164.9, 133.4, 133.0, 129.6, 129.5, 129.1, 128.6, 128.4, 128.3, 128.14, 128.08, 101.0, 99.3, 86.7, 76.1, 73.7, 72.8, 71.5, 70.7, 70.5, 69.5, 68.5, 68.2, 66.82, 66.77, 61.1, 20.6, 20.39, 20.35, 20.3.

MS (EI): *m*/*z* (%) = 874 [M^+] (0.61), 331 (100), 169 (100), 105 (100), 77 (100), 43 (100).

Anal. Calcd for $C_{45}H_{46}O_{18}$: C, 61.78; H, 5.30. Found: C, 61.78; H, 5.48.

Conditions B: According to the Typical Procedure for Conditions B, the reaction of CuI (0.0961 g, 0.5 mmol), $(HCHO)_n$ (0.0745 g, 2.5 mmol), **1l** (0.8610 g, 1.0 mmol), and Cy_2NH (0.36 mL, *d* = 0.91 g/mL, 0.3276 g, 1.8 mmol) in 1,4-dioxane (3.0 mL) afforded **2l** after chromatography (silica gel, PE–EtOAc, 1.5:1).

Yield: 0.6928 g (79%).

^{1}H NMR (300 MHz, $CDCl_3$): δ = 8.00–7.88 (m, 4 H, ArH), 7.81 (d, *J* = 7.5 Hz, 2 H, ArH), 7.55–7.19 (m, 9 H, ArH), 5.92 (t, *J* = 9.6 Hz, 1 H), 5.53 (dd, *J* = 9.6, 8.1 Hz, 1 H), 5.48–5.37 (m, 2 H), 5.27 (dd, *J* = 10.5, 8.1 Hz, 1 H), 5.22–5.12 (m, 1 H), 5.05 (dd, *J* = 10.7, 3.2 Hz, 1 H), 4.94 (d, *J* = 7.8 Hz, 1 H), 4.80–4.68 (m, 2 H), 4.64 (d, *J* = 7.8 Hz, 1 H), 4.46–4.35 (m, 1 H), 4.30–4.18 (m, 1 H), 4.15–4.00 (m, 4 H), 3.94 (t, *J* = 6.5 Hz, 1 H), 3.88–3.76 (m, 1 H), 2.108 (s, 3 H, Me), 2.105 (s, 3 H, Me), 1.98 (s, 3 H, Me), 1.94 (s, 3 H, Me).

Pentakis-allenyl β-D-Glucopyranoside 2m

Conditions A: According to the Typical Procedure for Conditions A, the reaction of CuI (0.0964 g, 0.5 mmol), $(HCHO)_n$ (0.3755 g, 12.5 mmol), **1m** (0.3709 g, 1.0 mmol), and *i*-Pr_2NH (1.27 mL, *d* = 0.718 g/mL, 0.9119 g, 9.0 mmol) in 1,4-dioxane (7.5 mL) afforded **2m** as a liquid after chromatography (silica gel, PE–EtOAc, 10:1).

Yield: 0.2570 g (58%); $[\alpha]_D^{20}$ +3.8 (*c* = 1.18, EtOAc).

IR (neat): 2923, 2869, 1956, 1730, 1632, 1605, 1459, 1438, 1361, 1310, 1261, 1233, 1079 cm^{-1}.

^{1}H NMR (300 MHz, $CDCl_3$): δ = 5.40–5.15 (m, 5 H), 4.94–4.70 (m, 10 H), 4.50–4.00 (m, 11 H), 3.73 (d, *J* = 10.8 Hz, 1 H), 3.63 (dd, *J* = 10.7, 4.1 Hz, 1 H), 3.47–3.17 (m, 4 H).

^{13}C NMR (75 Hz, $CDCl_3$): δ = 209.34, 209.29, 209.2, 209.1, 102.2, 88.3, 88.1, 87.9, 87.5, 87.4, 84.0, 81.4, 77.1, 75.9, 75.6, 75.5, 74.5, 71.1, 70.5, 70.3, 69.2, 68.4, 67.1.

ESI-MS: *m*/*z* = 463 [M^+ + Na], 458 [M^+ + NH_4].

Anal. Calcd for $C_{26}H_{32}O_6$: C, 70.89; H, 7.32. Found: C, 70.97; H, 7.37.

Conditions B: According to the Typical Procedure for Conditions B, the reaction of CuI (0.4789 g, 2.5 mmol), $(HCHO)_n$ (0.3746 g, 12.5 mmol), **1m** (0.3716 g, 1.0 mmol), and Cy_2NH (1.79 mL, *d* = 0.91 g/mL, 1.629 g, 9.0 mmol) in 1,4-dioxane (7.5 mL) afforded **2m** after chromatography (silica gel, PE–EtOAc, 10:1).

Yield: 0.2679 g (61%).

^{1}H NMR (300 MHz, $CDCl_3$): δ = 5.40–5.15 (m, 5 H), 4.97–4.70 (m, 10 H), 4.46–4.00 (m, 11 H), 3.72 (d, *J* = 10.5 Hz, 1 H), 3.63 (dd, *J* = 11.1, 3.6 Hz, 1 H), 3.45–3.17 (m, 4 H).

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Supporting Information

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(11) Crystal data for compound **2e**: $C_{19}H_{26}O_{10}$, MW = 414.40, orthorhombic, space group *P 21 21 21*, final *R* indices [$I > 2\sigma(I)$], $R1 = 0.0621$, $wR2 = 0.1540$, *R* indices (all data) $R1 = 0.1158$, $wR2 = 0.1877$, $a = 7.2505(5)$ Å, $b = 14.8053(12)$ Å, $c = 20.2383(12)$ Å, $V = 2172.5(3)$ Å^3, $T = 293(2)$ K, $Z = 4$, reflections collected/unique 15465/2934 ($R_{int} = 0.0561$), number of observations [$>2\sigma(I)$] 1690, parameters: 267. CCDC 999744 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

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